

Technology Stack

Introduction

The Smart Lender project uses Machine Learning and Web Technologies to predict loan eligibility. The selected technology stack provides efficient data processing, model training, prediction, and an interactive web interface for users.

Technology Stack Overview

Layer	Technology	Purpose
Programming Language	Python	Machine Learning and Backend Development
Frontend	HTML5, CSS3	User Interface
Backend Framework	Flask	Web Application Development
Machine Learning	Scikit-learn	Model Training and Prediction
Data Processing	Pandas	Data Loading and Manipulation
Numerical Computing	NumPy	Mathematical Operations
Data Visualization	Matplotlib, Seaborn	Exploratory Data Analysis
Dataset Balancing	SMOTE (Imbalanced-learn)	Handle Class Imbalance
Model Storage	Pickle	Save and Load Trained Model
IDE	Visual Studio Code	Development Environment
Version Control	Git & GitHub	Source Code Management

Frontend Technologies

HTML5

- Creates the structure of web pages.
- Develops the Home, Predict, and Result pages.

CSS3

- Enhances the appearance of the application.
- Provides responsive and user-friendly layouts.

Backend Technology

Flask

Flask is a lightweight Python web framework used to:

- Handle routing.
 - Process form data.
 - Load the trained model.
 - Perform loan predictions.
 - Display prediction results.
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Machine Learning Technologies

Scikit-learn

Used for:

- Data preprocessing
- Model training
- Model evaluation
- Cross-validation
- Prediction

Algorithms implemented:

- Decision Tree Classifier
 - Random Forest Classifier
 - K-Nearest Neighbors (KNN)
 - Gradient Boosting Classifier
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Data Processing Libraries

Pandas

- Reads and processes the loan dataset.
- Cleans and transforms data.

NumPy

- Performs numerical computations.
 - Handles arrays and mathematical operations.
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Data Visualization

Matplotlib

Creates charts for dataset analysis.

Seaborn

Generates statistical visualizations such as:

- Count plots
 - Distribution plots
 - Swarm plots
 - Categorical plots
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Model Persistence

Pickle

Used to:

- Save the trained Random Forest model.
 - Load the model into the Flask application for real-time prediction.
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Development Tools

Visual Studio Code

- Code editor for Python, HTML, and CSS.
- Supports debugging and project management.

Git & GitHub

- Version control system.
 - Repository hosting.
 - Collaboration and project tracking.
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System Workflow

Dataset



Data Preprocessing



Feature Engineering



Model Training



Model Evaluation



Best Model Selection



Model Saved (.pkl)



Flask Web Application



Loan Prediction Result

Advantages of the Technology Stack

- **Easy to develop and maintain.**
 - **Fast model training and prediction.**
 - **Lightweight Flask framework.**
 - **Scalable architecture.**
 - **Open-source technologies.**
 - **Easy deployment.**
 - **Strong community support.**
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Conclusion

The Smart Lender project leverages a robust and modern technology stack that integrates machine learning, web development, and data processing tools. This combination enables accurate loan eligibility prediction while providing a responsive and user-friendly web application.